Test Cases and Test Report

Project: E-Commerce & Order Management System (Olist) · **Group:** TarushTanishkTanishq **Test lead:** Tanishq Mahajan · **Testers:** all members · **Version tested:** commit `427aeb6` (main) + test scripts **Test execution date:** 15 October 2026

1. Summary

Total test cases	Passed	Passed with note	Failed	Not run
31	29	2	0	0

The two "passed with note" cases (TC-02, TC-24) behaved correctly. Their observed numbers differ from the expected figures only because the test machine used an older copy of the public Olist dataset (see §2 and the notes in the tables). Both are scheduled for a confirmation re-run on the team laptop with the current Kaggle files.

2. Test environment

Item	Value
OS	Ubuntu 24.04 (Linux x86-64)
Database	PostgreSQL 16.15, `pgcrypto`
Runtime	Python 3.13, Flask 3.1, scikit-learn 1.9, pandas 3.0, psycopg 3 (pure-Python build)
Browser	Headless Chromium (Playwright) for UI screenshots
Data	Olist Brazilian E-Commerce dataset, 9 CSVs (older public copy: `olist_order_reviews_dataset.csv` has 100,000 rows; the current Kaggle release has 99,224)
Not installed	LightGBM (optional, per README); the model comparison ran without it

3. How the tests were run

Level	Script / tool	Evidence file
Build	`psql -f db/00_run_all.sql` (same as `setup_db.bat`)	`logs/TC01_setup_db.log`
ML	`python ml/train.py` (same as `train_model.bat`)	`logs/TC29_train_model.log`
Database unit tests	`psql -f db/15_test_suite.sql`; every test runs in `BEGIN … ROLLBACK`	`logs/TC02-TC21_db_test_suite.log`
Integration / system	`python tests/api_tests.py` against the running app (Front end → API → SQL → DB → result)	`logs/TC15-TC31_api_tests.log`
UI evidence	Browser screenshots of the pages under test	`screenshots/*.png`

Priority: H = high, M = medium, L = low.

4. Test cases and results

4.1 Database build and data

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-01	NFR-7	Build the database on an empty server	All 8 steps finish, no errors	All 8 steps finished, exit code 0, 23.7 s	H	Pass
TC-02	Data	Compare table row counts with the CSVs	orders 99,441 · order_item 112,650 · payment 103,886 · review 99,224	orders 99,441 · order_item 112,650 · payment 103,886 · review **100,000** (+ 27 states, 19,177 zips, 3,095 sellers, 32,951 products)	H	Pass with note: review count equals its source file (100,000 rows in the older copy). Re-check on the current Kaggle file
TC-03	NFR-3	INSERT `order_item` with unknown `product_id`	FK violation, row rejected	`violates foreign key constraint "order_item_product_id_fkey"`	H	Pass
TC-04	NFR-3	INSERT `payment` with value −5	CHECK violation	`violates check constraint "payment_payment_value_check"`	M	Pass
TC-05	Data	`SELECT * FROM v_data_quality`	Anomalies listed	1,381 rows: 775 orders without items, 249 payment ≠ items total, 167 zips without coordinates, 166 carrier date before purchase, 23 delivered before carrier, 1 order without payment	L	Pass

4.2 Business rules (triggers and procedures)

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-06	FR-6	UPDATE a delivered order to `shipped`	Error, status unchanged	`Order 00010242… is already delivered and cannot change to shipped`	H	Pass
TC-07	FR-6	UPDATE an `approved` order to `shipped`	Allowed	Row updated, status = shipped	H	Pass
TC-08	FR-7	Same change as TC-07	Row in `order_status_log`	Log row: old `approved`, new `shipped`, changed_by `postgres`	H	Pass
TC-09	FR-8	DELETE a delivered order	Error	`Order 00010242… was delivered and cannot be deleted`	H	Pass
TC-10	FR-9	INSERT review for an order in `processing`	Error	`Order d3c8851a… is still "processing": it can be reviewed only after delivery`	M	Pass
TC-11	FR-5	`CALL sp_place_order` (valid customer, 2 items)	Order + 2 items + payment; stock reduced	1 order, 2 items, 1 payment row; stock 170→169 and 50→49	H	Pass

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-12	FR-5	`sp_place_order` with unknown customer	Error, nothing inserted	`Unknown customer 0000…`; orders 99,441 before and after	H	Pass
TC-13	FR-10	Order 2 units of a product with stock 1	CHECK fails, whole order rolled back	`violates check constraint "product_stock_qty_check"`; stock still 1, 0 orders added	H	Pass
TC-14	FR-5	`CALL sp_cancel_order` on a delivered order	Error	`Order 00010242… is already delivered, it cannot be canceled`	M	Pass

4.3 Security and RBAC

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-15	FR-1	Log in as `admin` with a wrong password	"Wrong username or password", no session	API: HTTP 401 "Wrong username or password". DB: `fn_login` returns 0 rows (1 row with the right password)	H	Pass
TC-16	FR-1	`SELECT password_hash FROM app_user` as `olist_app`	Permission denied	`permission denied for table app_user`	H	Pass
TC-17	FR-16	Log in as `seller`, open Orders	Only the seller's orders	Seller sees 1,854 of 99,441 orders; 0 orders without its items (DB and UI)	H	Pass
TC-18	FR-16	As `olist_analyst`, `SELECT lat FROM zip_code`	Permission denied	zip/city readable; `lat` → `permission denied for table zip_code`	H	Pass
TC-19	FR-2	As `analyst`, delete a product	403, PostgreSQL error 42501	API: HTTP 403 "Your role (analyst) cannot do this" (the app's role check stops it first). DB directly: `42501: permission denied for table product`	H	Pass: both layers block it
TC-20	NFR-2	Search with `' OR 1=1 --`	Treated as literal; no extra rows	`order_id` filter → HTTP 400 "bad id"; `state` filter `SP' OR '1'='1` → 0 rows	H	Pass
TC-21	FR-20	SQL console: `DELETE FROM orders`	Rejected	Plain DELETE → HTTP 400. DELETE hidden in a CTE → rejected by the read-only transaction	H	Pass

4.4 Front end and reports

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-22	FR-3/4	Create, edit, delete a product as `manager`	Each succeeds; SQL panel shows INSERT/UPDATE/DELETE	All three succeeded; the response SQL log shows `INSERT INTO product…`, `UPDATE product SET stock_qty = 25…`, `DELETE FROM product WHERE…` (1 row)	H	Pass

ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-23	FR-11	Filter orders: state = SP, status = delivered	Counts update; rows match	50 rows on page 1, all SP + delivered; dropdown count delivered (SP) = 40,501	M	**Pass**
TC-24	FR-12	Run every report (Q1–Q16 + Q13b)	Rows returned, no error	All 17 ran without error; 16 returned rows. Q2 (delivered orders never reviewed) returned 0 rows	H	**Pass with note**: in the older data copy every order has a review, so 0 rows is the correct answer there. Re-check on the current file, which has orders without reviews
TC-25	FR-12	EXPLAIN and CSV export on Q5	Plan shown; CSV has same rows	28-line plan with Execution Time; CSV 76 rows = UI 76 rows	M	**Pass**
TC-26	FR-14	Open "ER model & schema"	ER diagram + live catalog	Diagram rendered; tabs Tables & keys, Constraints, Objects, Privileges, Normalization; 81 columns and 45 constraints read live	H	**Pass**
TC-27	FR-15	Transactions lab: lost update without / with `FOR UPDATE`	First loses one update; second keeps both	Without lock: stock 170 → 169 (one update lost). With `FOR UPDATE`: 170 → 168 (both kept)	H	**Pass**
TC-28	NFR-5	Order search on the full dataset	< 1 s	18, 15, 474 (with facet counts), 31 ms	M	**Pass**

4.5 ML component

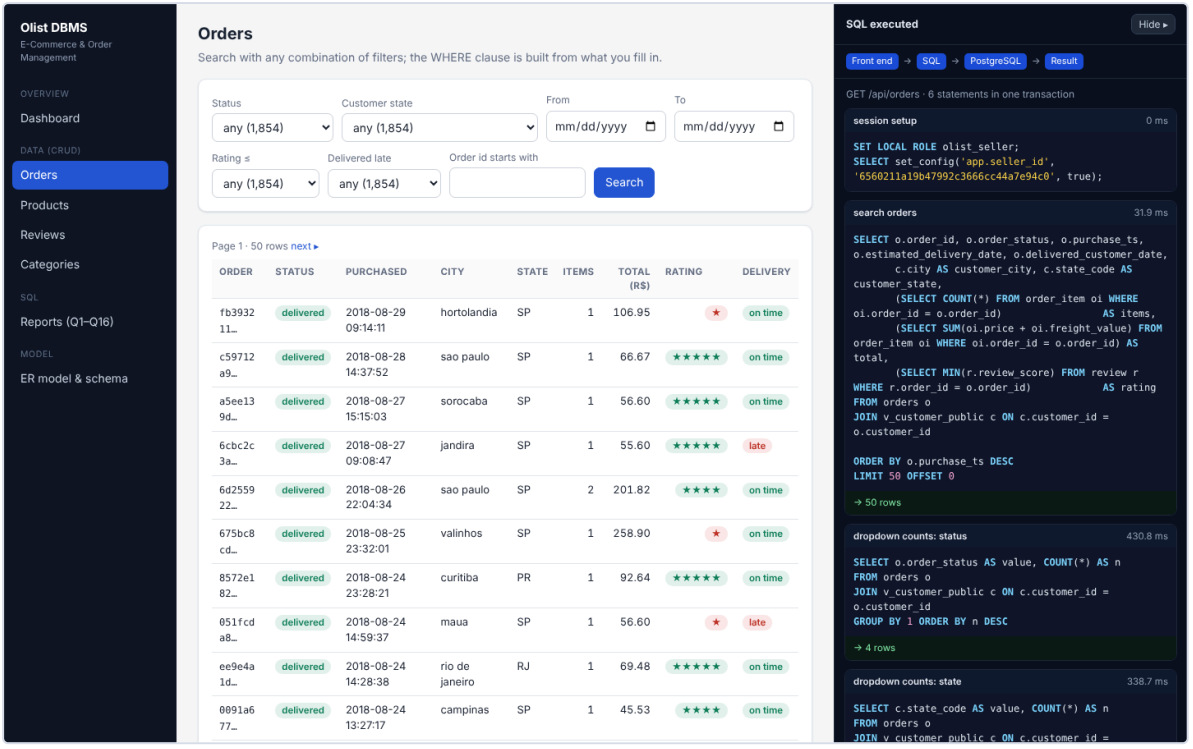
ID	Req.	Input / action	Expected result	Actual result	Pri	Status
TC-29	NFR-10	Train the model	Models compared with CV; best saved; metrics in `ml_model`	4 models compared with 3-fold time-series CV; Logistic Regression selected (CV ROC-AUC 0.647). Test Jun–Aug 2018: ROC-AUC 0.721, PR-AUC 0.112, recall 0.664; delivery-days MAE 3.51 vs 12.72 for Olist's promise; 98,666 predictions written	H	**Pass**
TC-30	FR-17	Place an order in the UI	Risk + predicted days; row in `ml_prediction`	Risk "medium", p(late) 0.383, 10.9 days predicted vs 25 promised; `INSERT INTO ml_prediction` in the SQL log	H	**Pass**
TC-31	FR-18	"Predict risk only"	Prediction shown; order not saved	Prediction returned; order count 99,442 before and after (rolled back)	M	**Pass**

5. Defect log

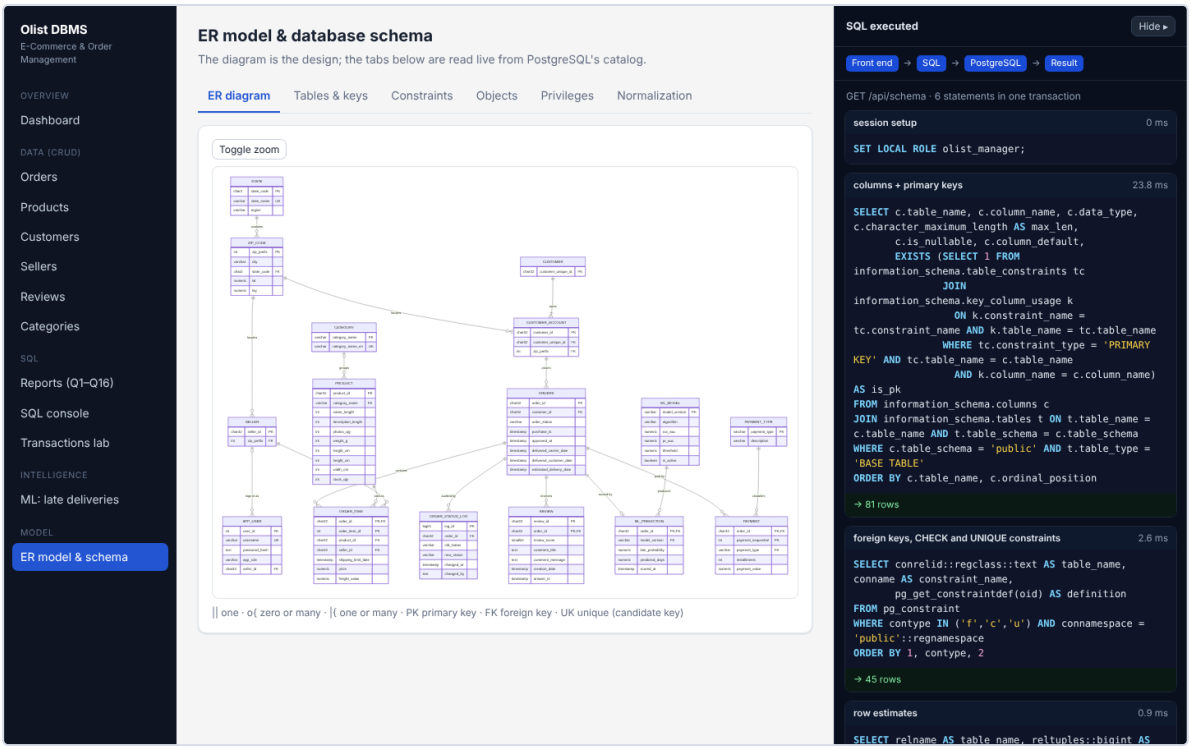
ID	Found in	Description	Severity	Fix	Status
D-01	System test	`http://localhost:5000/` returned 404	Medium	Added index route in `backend/app.py`	Fixed (29 Sep)
D-02	Installation test	Login timed out without a clear message when PostgreSQL was not running	Low	`run_app.bat` checks `pg_isready` first	Fixed (29 Sep)
D-03	Test-plan review	TC-19 expected only a PostgreSQL error, but the API blocks the action earlier with its own role check	Low (documentation)	Expected result updated: both layers block the action	Closed
D-04	Test run	An apostrophe in an `\echo` in `db/13_index_demo.sql` broke the first heading of the index demo	Low	Removed the apostrophe	Fixed (15 Oct)

No functional defects were found in this test run.

6. Evidence



TC-17: logged in as `seller`, every dropdown shows 1,854 orders; the SQL panel shows `SET LOCAL ROLE olist_seller` and the RLS setting.



TC-26: ER model and live schema page, with the catalog queries in the SQL panel.

All logs and screenshots are in `05_Testing/Test_Evidence/`.